# MinION Loading

# How much concentration do I really need?

- Final pool
  - Need 100-150 fmol to load the flow cell
  - = 11ul of 10-15nM

- Individual samples:
  - Ideally individual samples would also have 10-15nM
    - (7-10ng/ul for 1000bp library)
    - (13-20ng/ul for 2000bp library)
  - Total mass (ng) per sample added to pool will depend on:
    - library size
    - # of samples in the pool

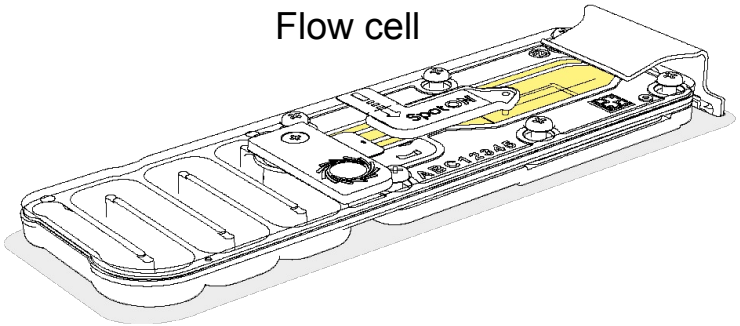
# Before Loading...

- Thaw loading reagents
  - ~30 min at room temperature
- Take out SpotON flow cell from 4◦C
- If it's your first time loading, watch this video to see how it is done
- Ensure that the MinION is connected to the computer and the software (MinKNOW) is up to date
- Insert the flow-cell in the instrument
- Perform a flow cell check and ensure that at least 800 pores are available* (ideally 1200-1800)

MinION instrument

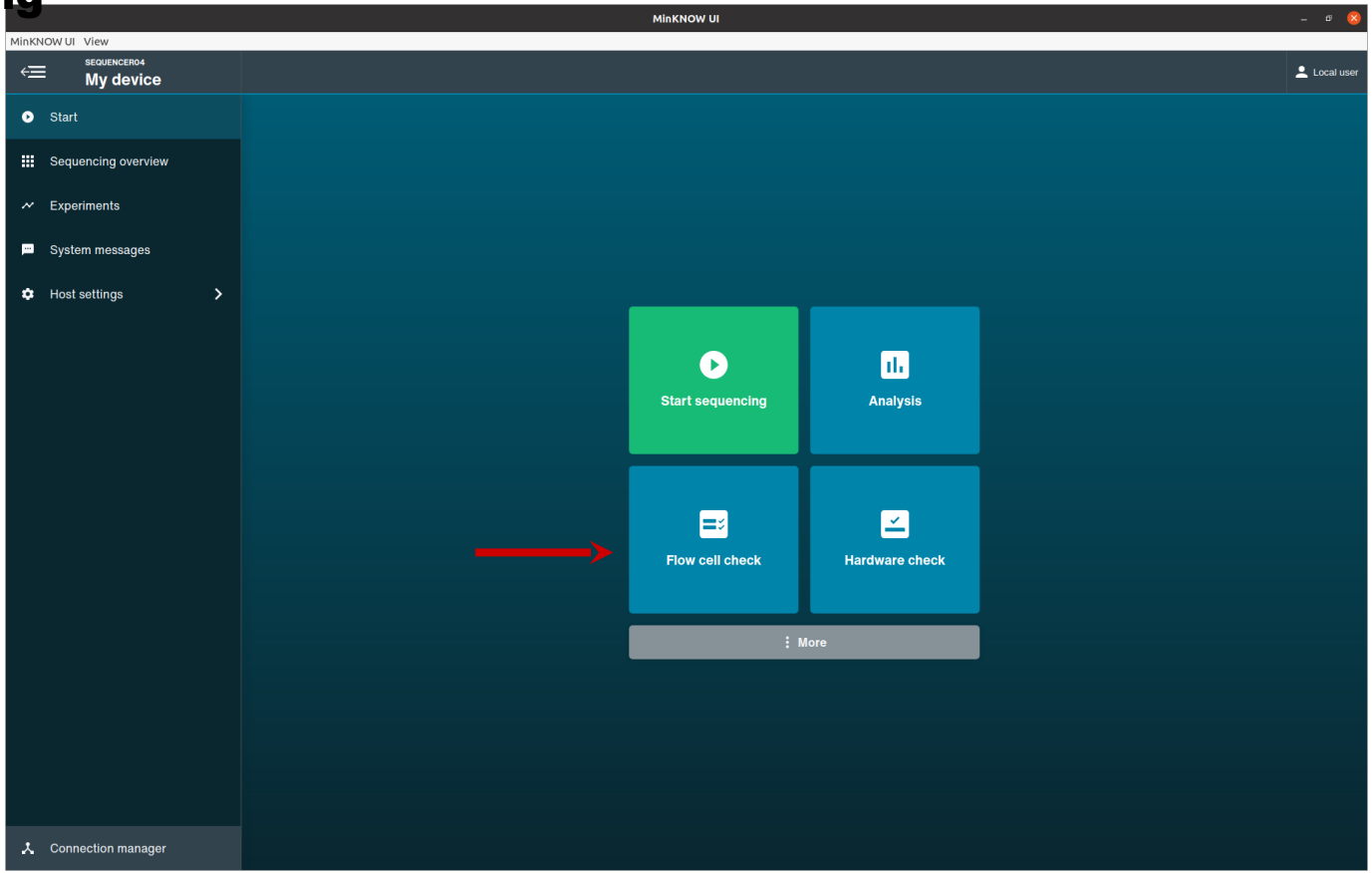


Flow cell



*ONT guarantees 800 active pores; you can sequence with fewer, but less data will be generated

# Perform a Flow cell check to identify the number of available pores for sequencing



# Loading the flow cell

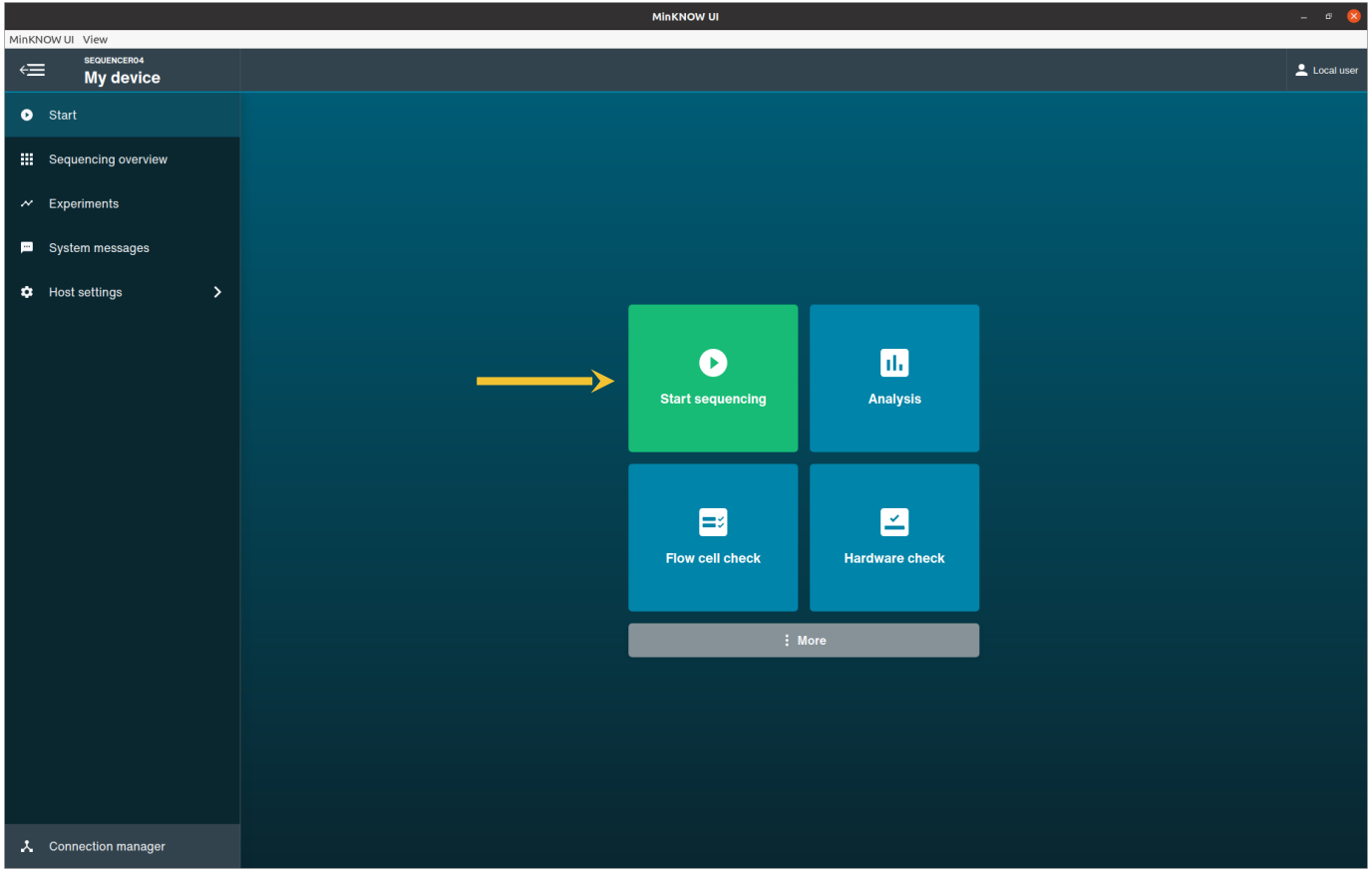
- Follow the ONT protocol for “Priming and loading the MinION and GridION Flow Cell” (link here)

- If it is your first time loading, watch the “Priming and loading your flow cell” video (link here)

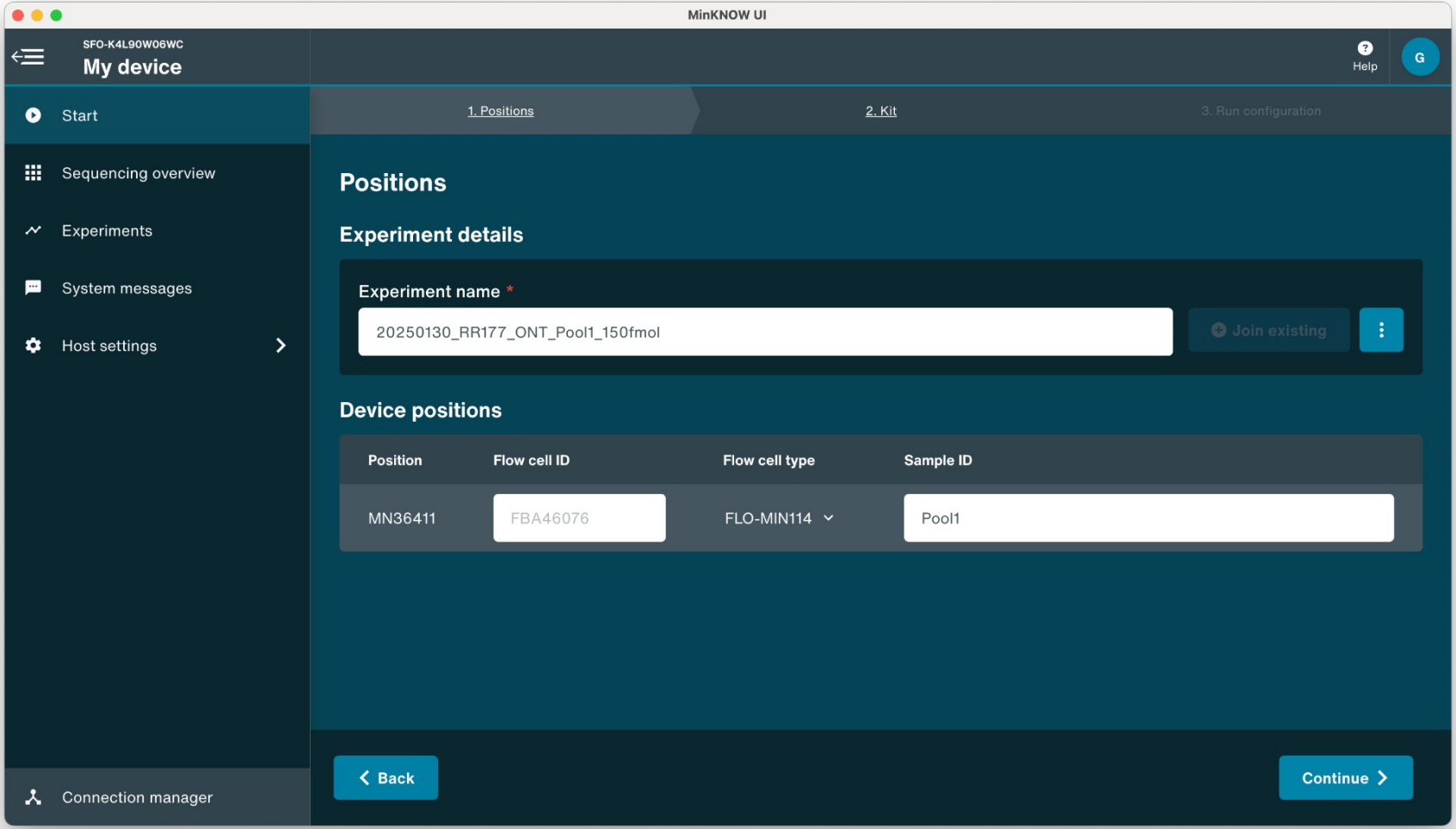
**Notes:**

- Generally, library beads (LIB) are used for loading, and library solution (LIS) is not used
- Use Bovine Serum Albumin (BSA) if you have the exact catalog number listed in the protocol (otherwise skip)

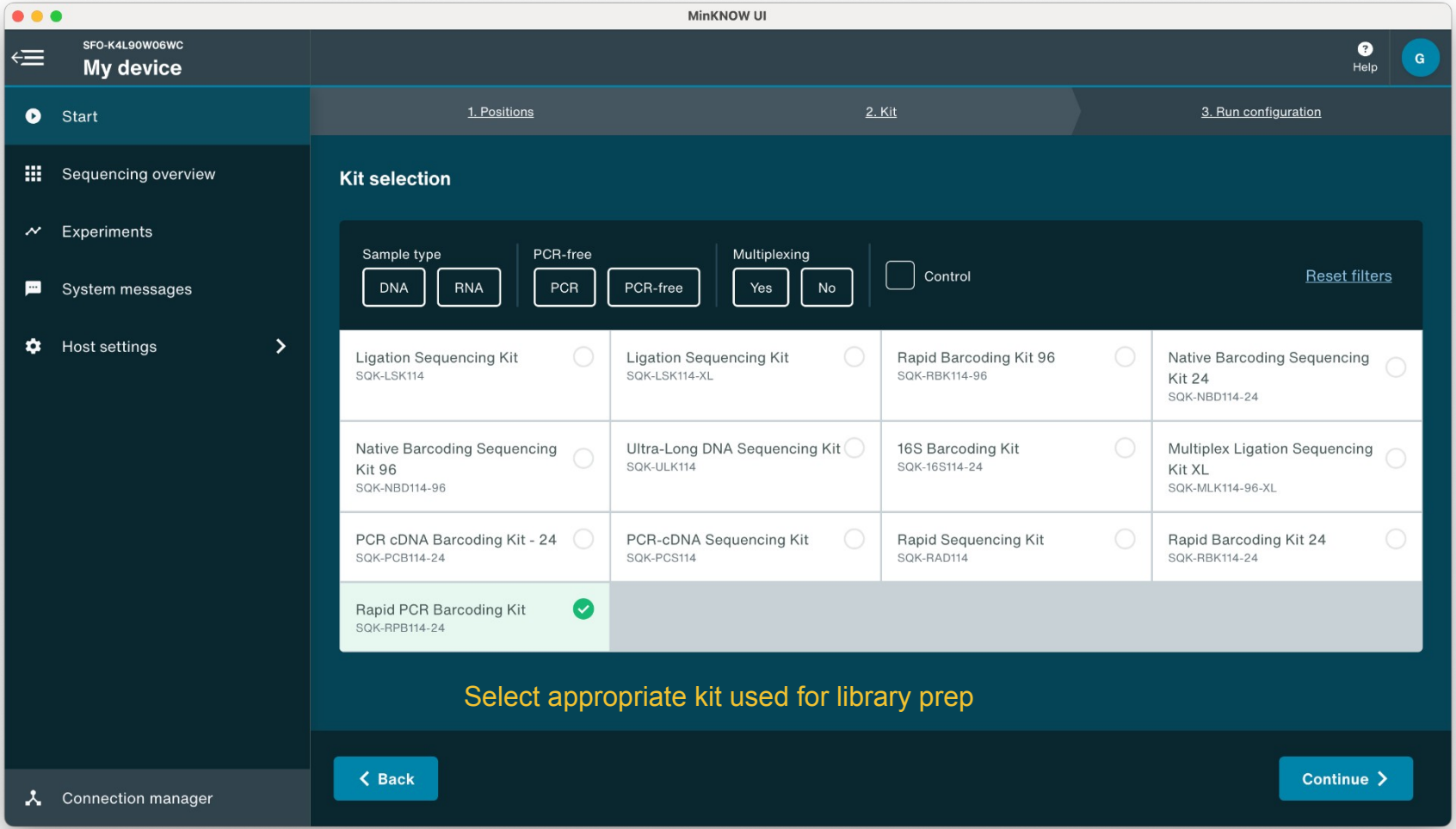
# Starting a sequencing run



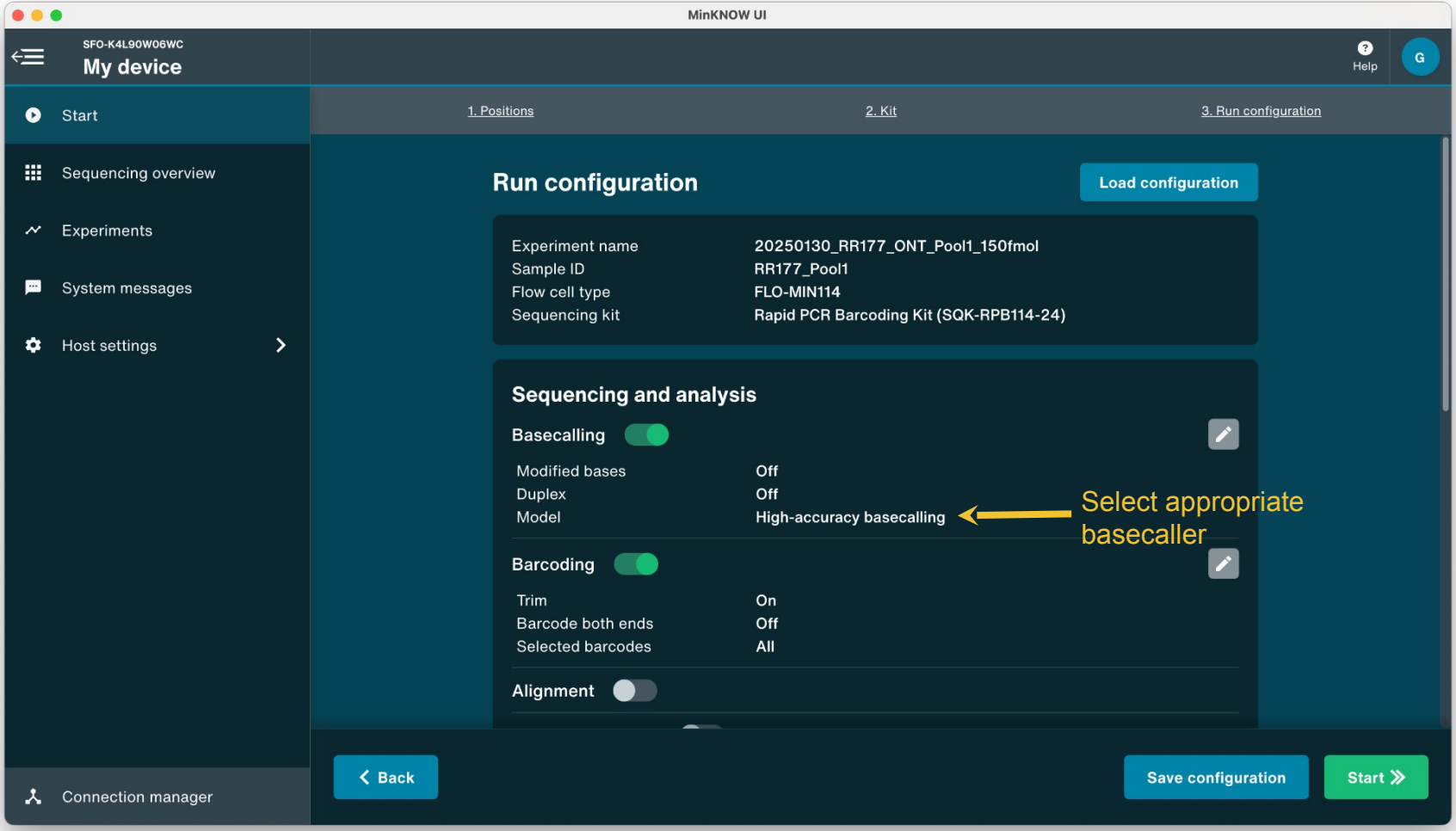
# Starting a sequencing run



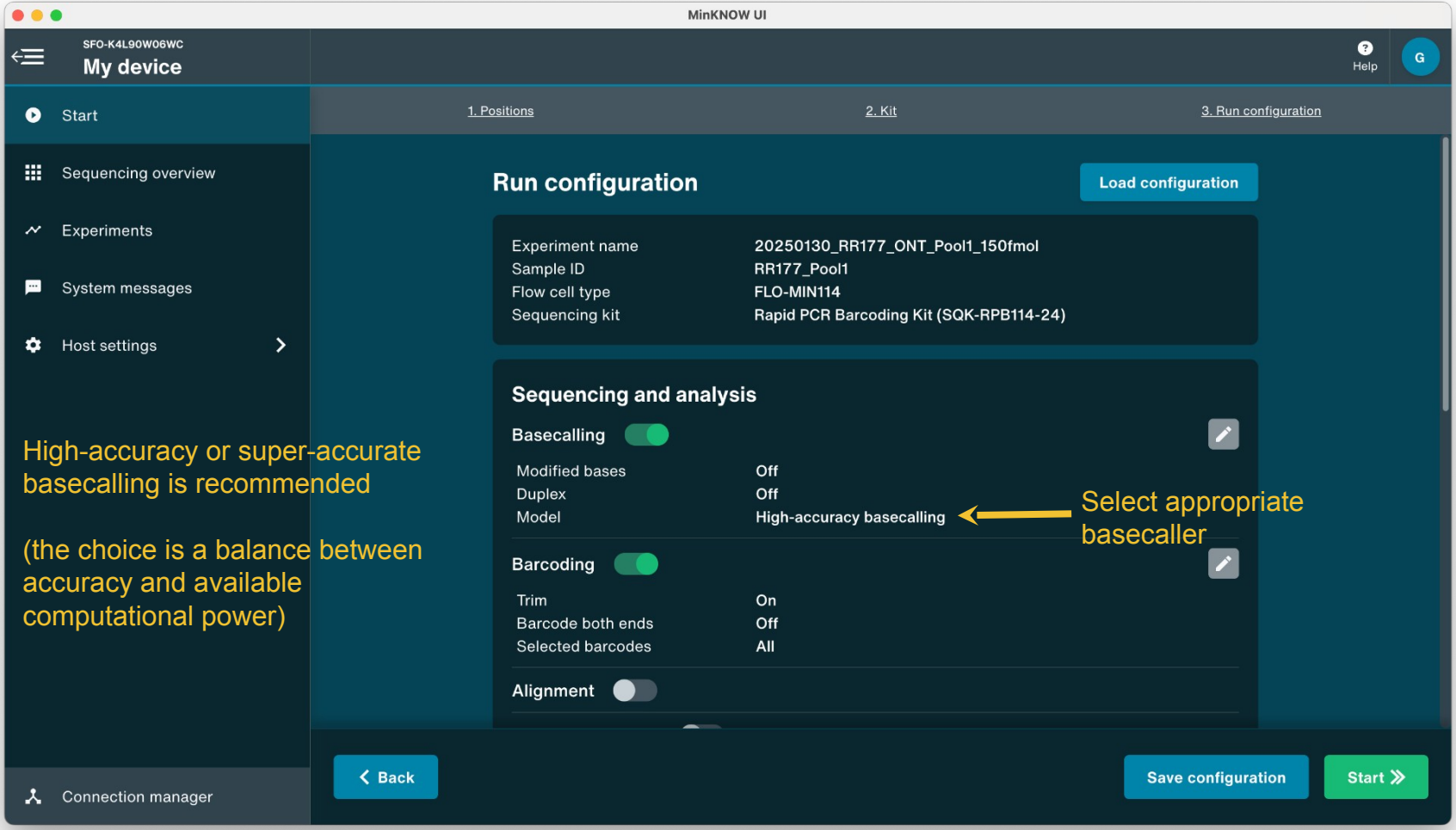
# Select library prep kit used: SQK-RPB114-24 Rapid PCR Barcoding Kit



# Select a basecaller



# Select a basecaller



# Intro to basecallers

What do they do?

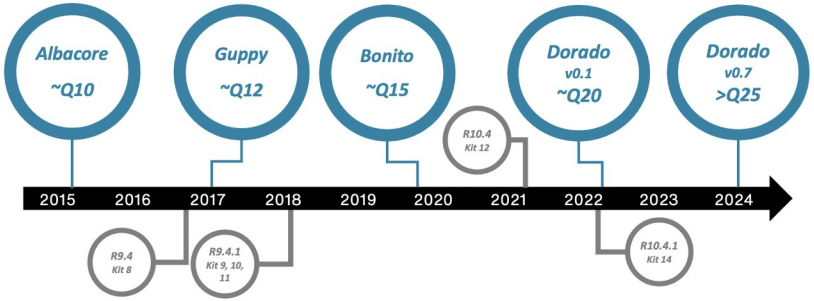
- Input raw ONT data (POD5)
- Output basecalled data (FASTQ)

Basecaller options

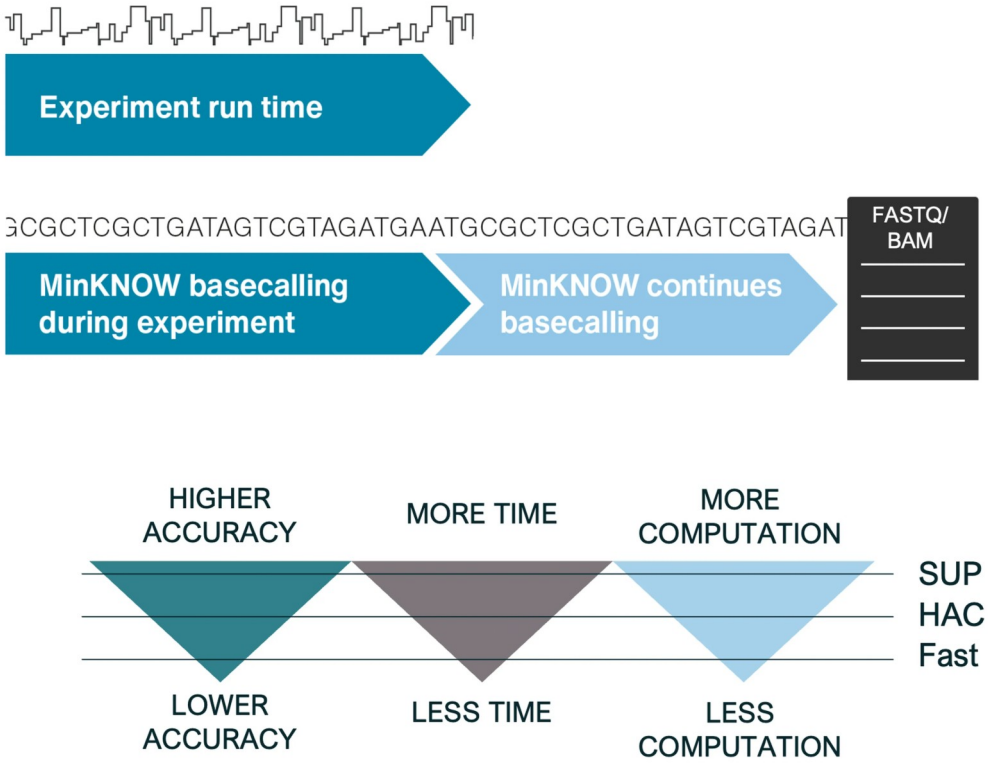
- Fast
- High Accuracy (HAC)
- Super Accuracy (SUP)

ONT Basecaller programs

*ONT periodically releases new basecallers*



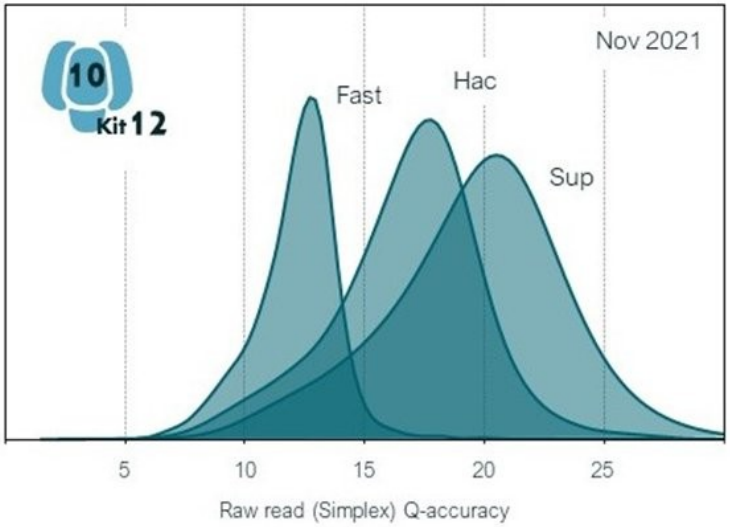
# Basecallers differ in accuracy, computational power required, and time




Higher accuracy basecallers will require more computation and time

Fast basecallers can keep up with sequencing runs

# Comparison of Dorado basecalling models - accuracy




Basecalling accuracy is measured in Phred quality score (Q score)

A Phred quality score (Q score) is a measure of confidence of the basecall based on the estimated error rate.

| Phred Quality Score | Probability of incorrect base call | Base call accuracy |
|---|---|---|
| 10 | 1 in 10 | 90% |
| 20 | 1 in 100 | 99% |
| 30 | 1 in 1000 | 99.9% |
| 40 | 1 in 10,000 | 99.99% |

# The choice of basecaller is dependent on the computational resources available, time, and accuracy required.

**Fast basecalling:** fastest, least computationally intense. Compatible with real-time basecalling on all nanopore devices with compute. Recommended for quick, real-time insights on sequencing data when compute resources are limited.

**High accuracy basecalling (HAC):** highly accurate, intermediate speed and computational requirement. Recommended for high-throughput projects focusing on variant analysis.

**Super accuracy basecalling (SUP):** the most accurate and computationally intense. Recommended for de novo assembly projects and low-frequency variant analysis (e.g. somatic variation and single-cell applications).

https://nanoporetech.com/platform/accuracy

# So which basecaller should I use?

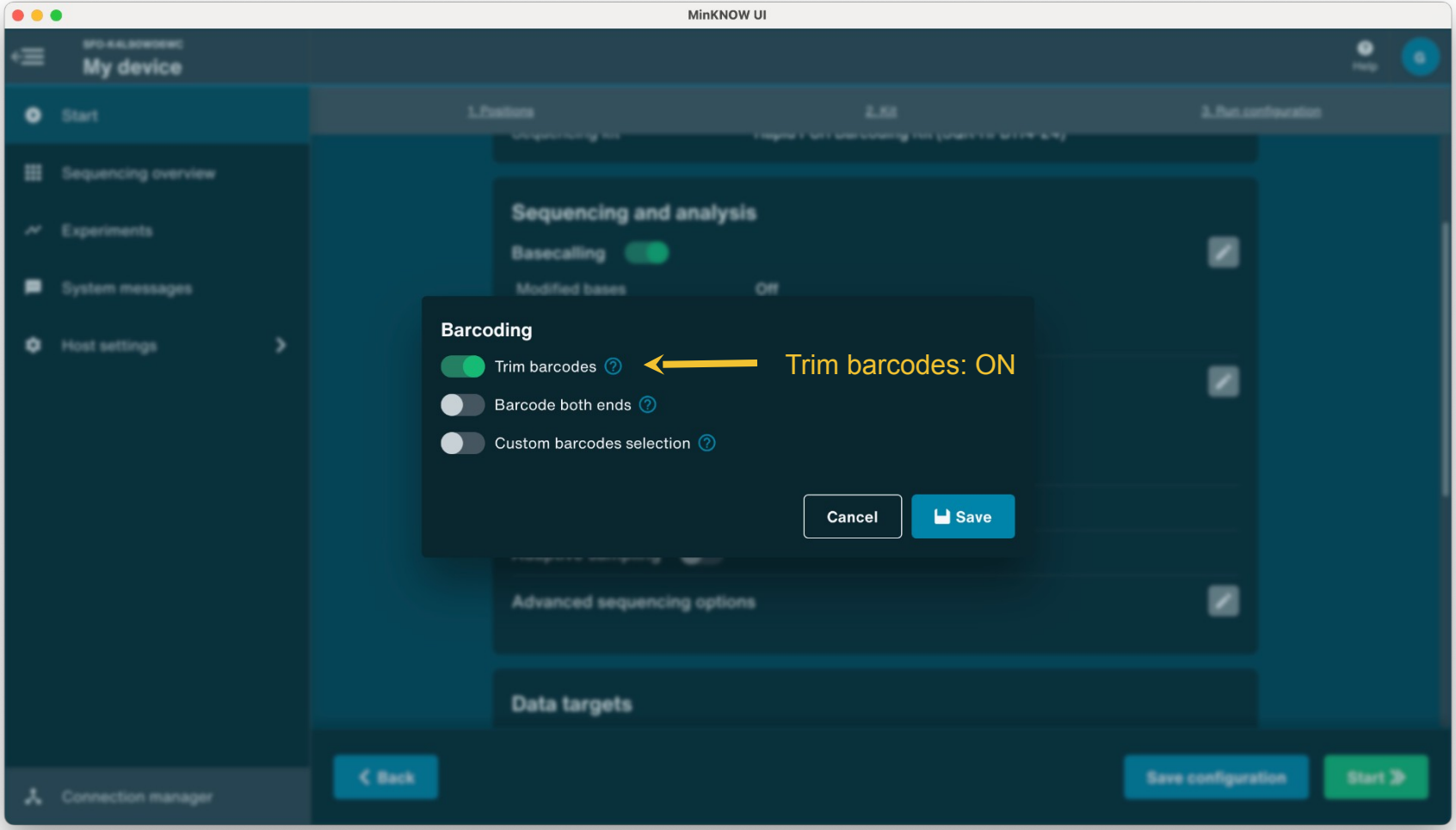
**Real time during the run:** HAC if possible, but if computation is limited, choose fast

**Preliminary data analysis:** fast is okay

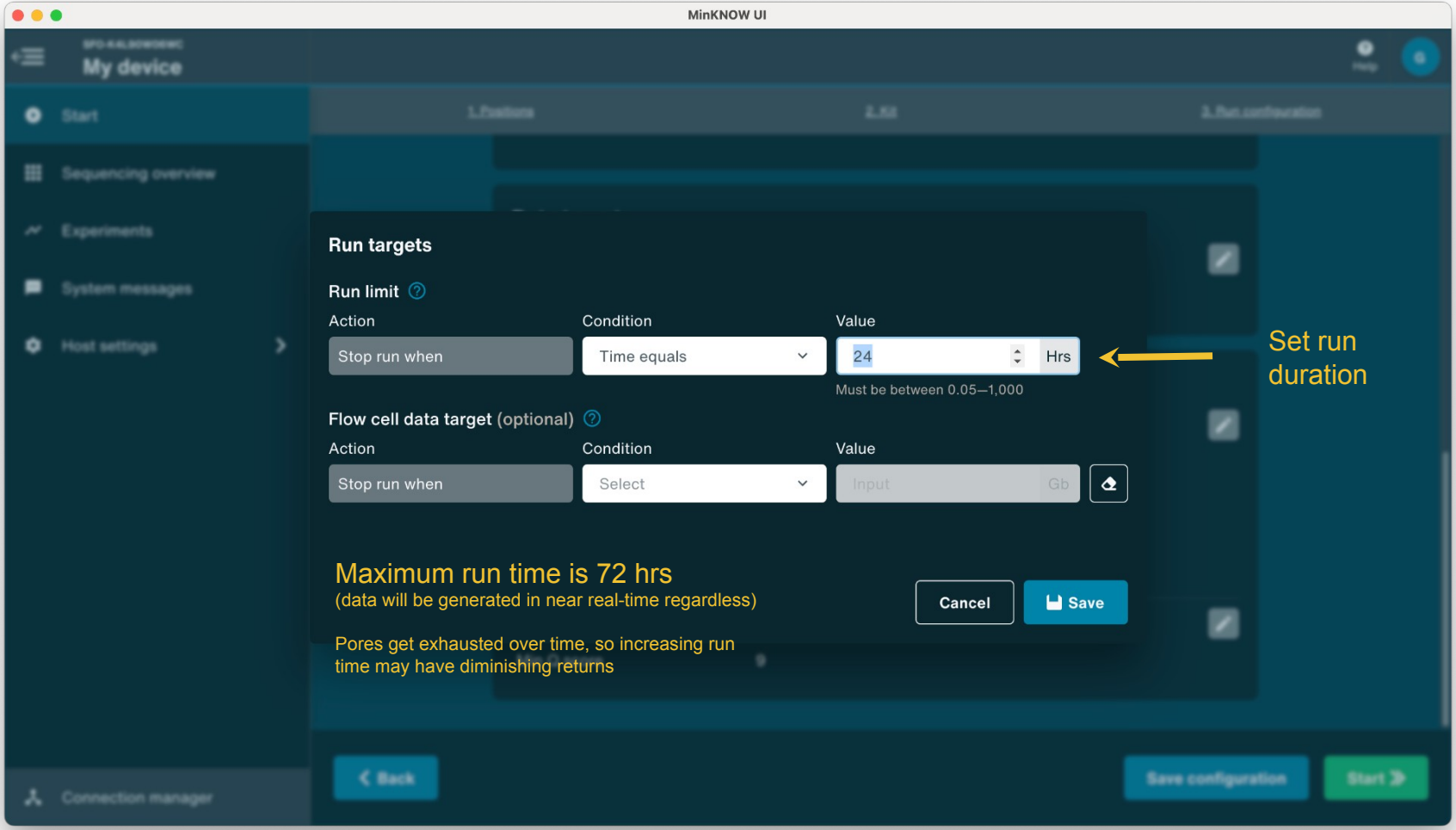
**Final data analysis:** HAC or SUP* is recommended

**SUP may be too computationally intensive*

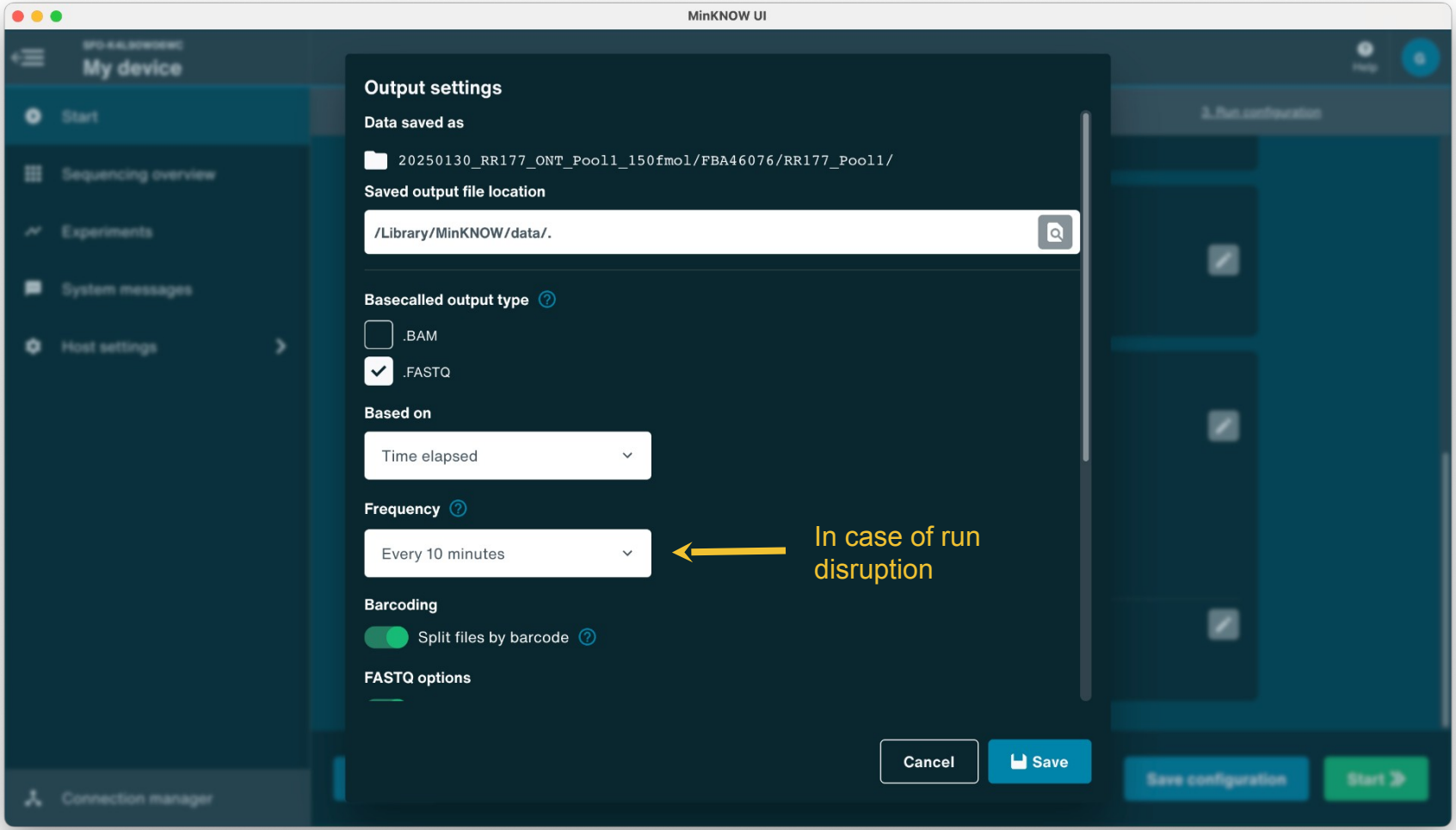
# Trim barcodes: ON



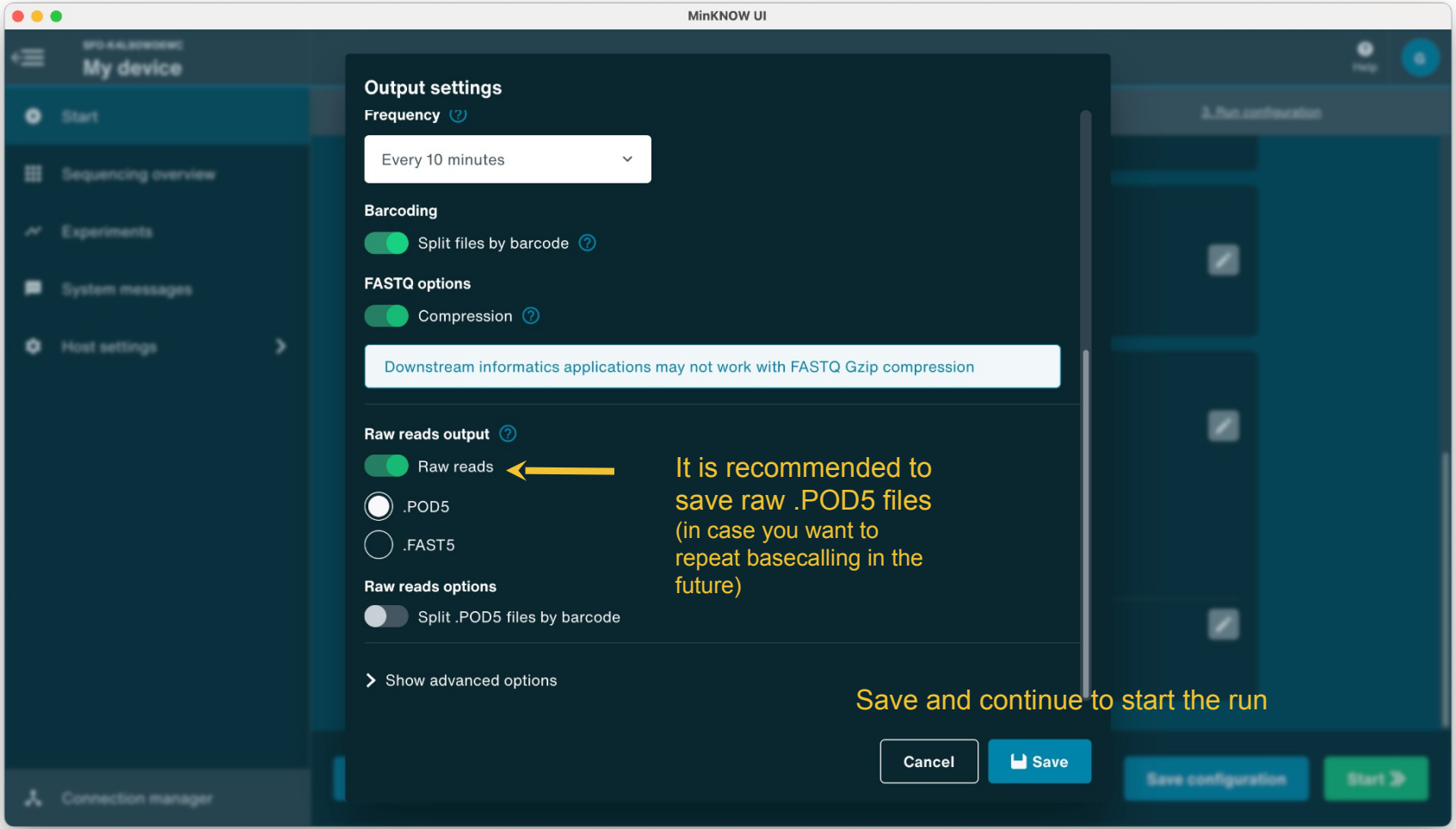
# Set run duration



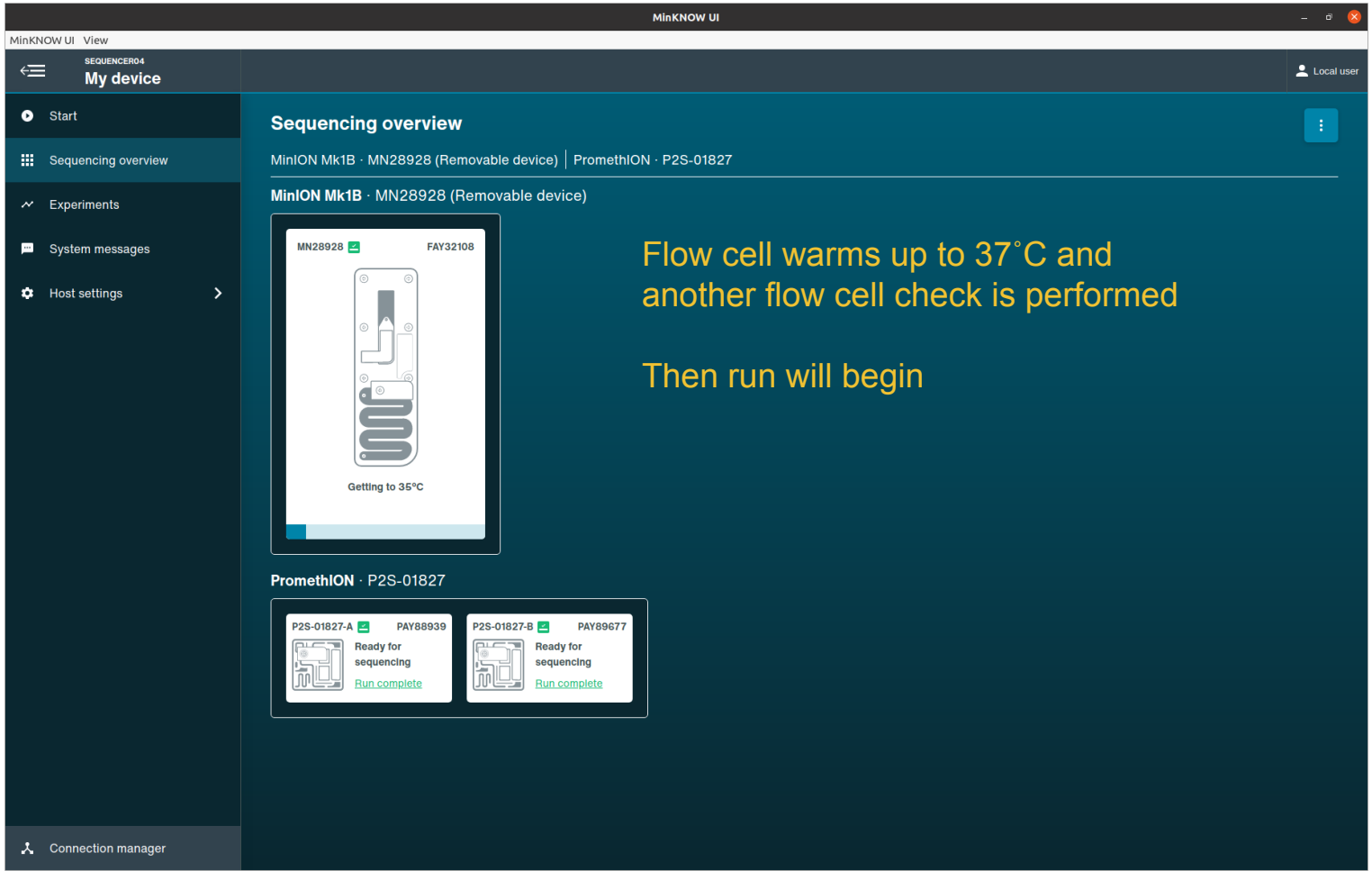
# Save output files every 10 minutes



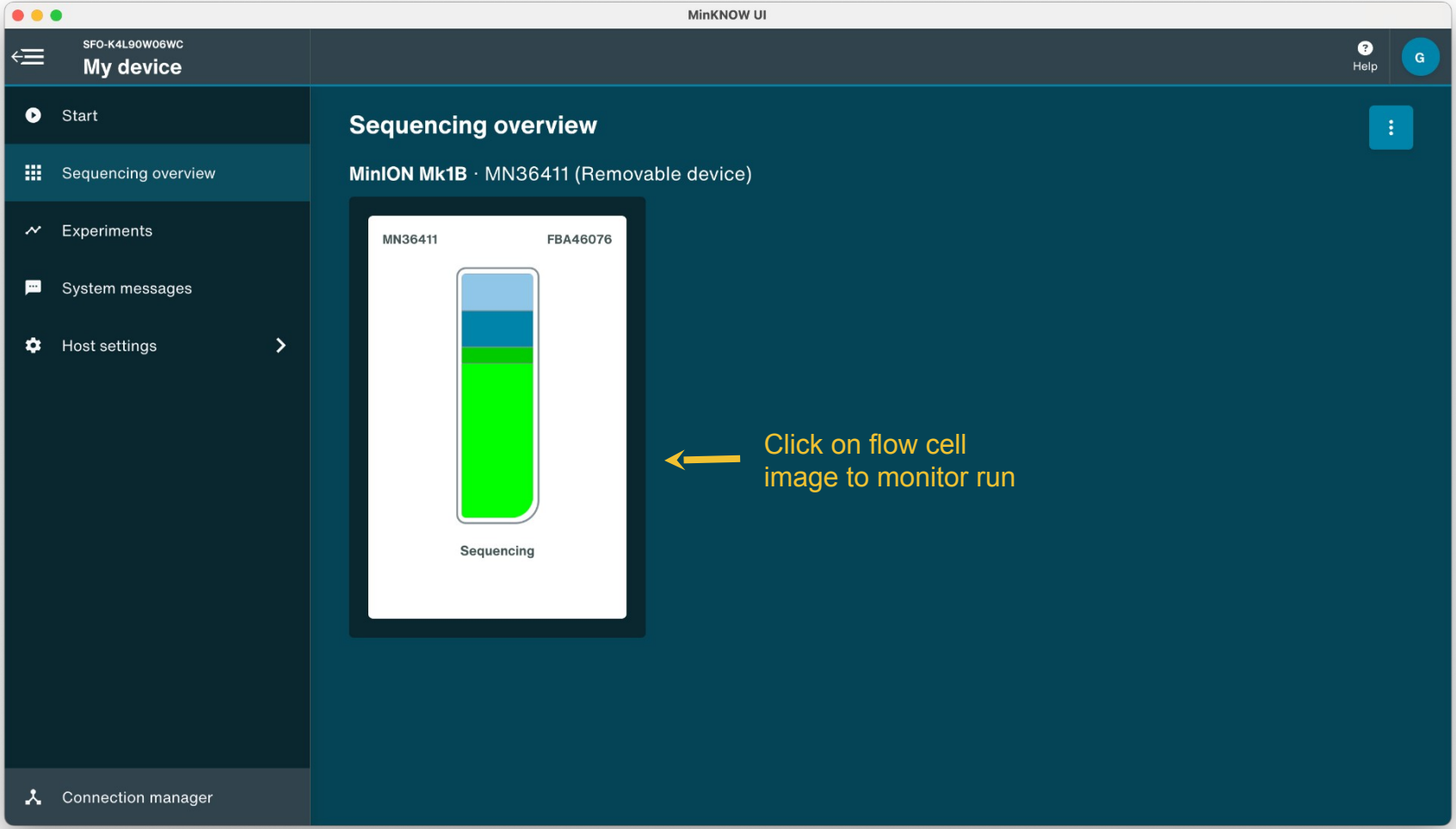
# Choose output files to store



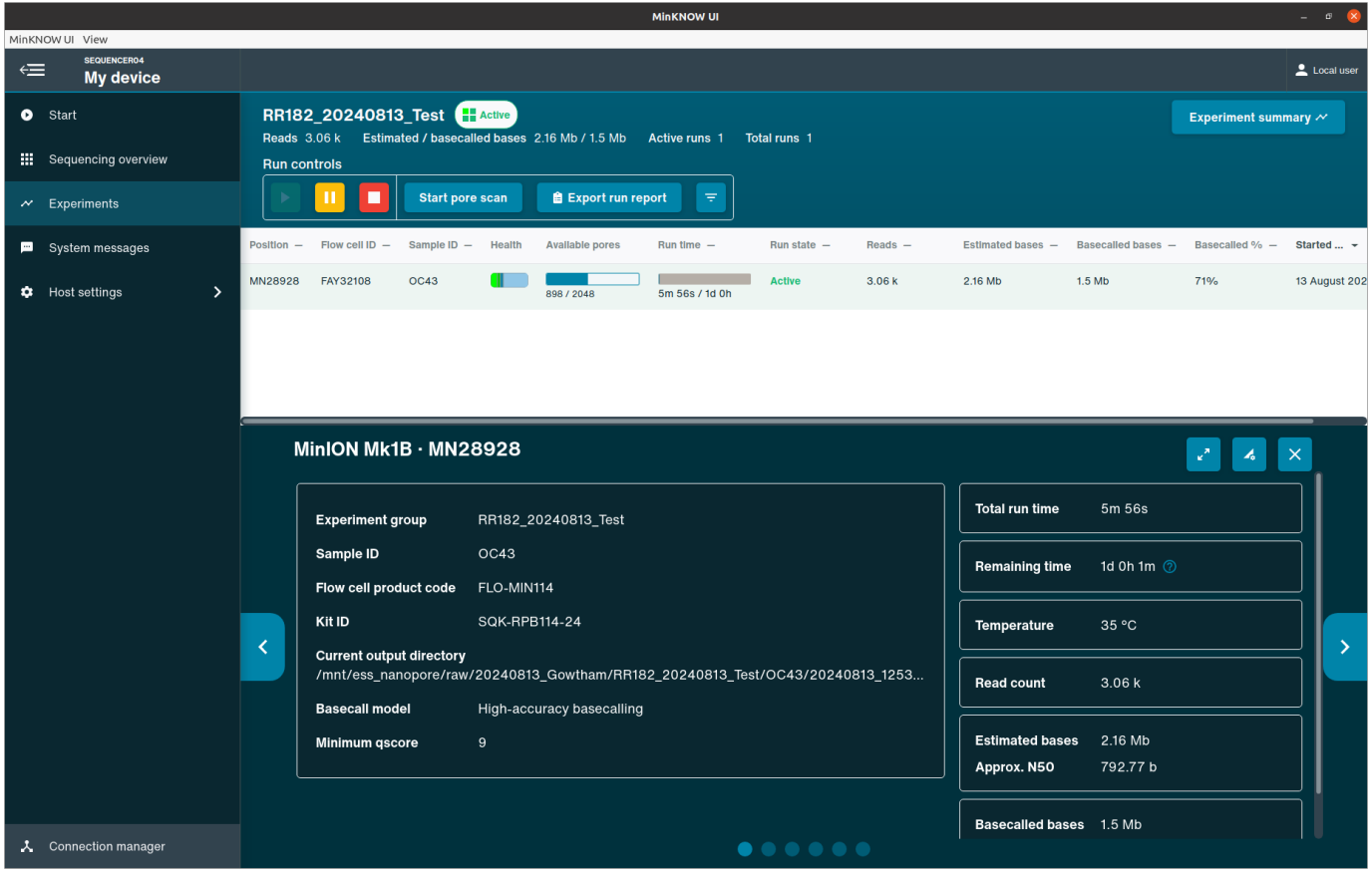
# Start the run



# Monitor the run



# Monitor the run



# After a run

1. Check the run report to see approximately how many pores are available

1. Wash the flow cell if it is worth saving

1. Run flow cell check to see how many pores are still available after wash

1. Start another run *or* add storage buffer and save at 4C

# Computer considerations

- POD5 files are very large
  - Ensure there is a storage plan in place to maintain space on the computer
  - You may want to move or discard POD5 files at a regular cadence

- Always have 300GB available before every run
  - A backup 5TB+ drive or a NAS (Network attached storage) are options to use for longer storage to keep your local storage clean

Next: 08-O_Sequencing Run QC for ONT  Slides & Activity